

AspectSense: Multi-Criteria Product Intelligence via Metadata-Gated Hybrid RAG

Technical Report

Abstract

Standard Retrieval-Augmented Generation (RAG) frameworks applied to large customer review corpora suffer from severe sampling bias and cross-product semantic drift. Naive top- k dense retrieval frequently surfaces isolated negative complaints, leading Large Language Models (LLMs) to hallucinate widespread hardware defects on products with overwhelmingly positive macro ratings. In this work, we propose **AspectSense**, a dual-stream intelligence engine that reconciles macro statistical realities with micro qualitative evidence. AspectSense couples an indexed relational SQL engine with dynamic metadata-gated hybrid retrieval—fusing dense multilingual semantic embeddings and sparse lexical BM25 scores via Reciprocal Rank Fusion (RRF), followed by cross-attention reranking using BAAI/bge-reranker-base. The retrieved context is synthesized using Gemma 3 4B into structured reports with automated citation verification. Across a rigorous benchmark of 50 multi-aspect evaluation test suites spanning 22 smartphone models, AspectSense achieves a **94.0% Hit Rate@5**, an **MRR@5 of 0.900**, and an **88.0% Aspect Precision@5**, outperforming standalone dense retrieval by 47.6% absolute precision while maintaining an interactive latency profile.

1 Problem Statement

Retrieval-Augmented Generation (RAG) has emerged as the standard paradigm for querying unstructured text. However, deploying naive RAG architectures on multi-market consumer product reviews reveals two systemic architectural vulnerabilities:

1. **Sampling Bias and Qualitative Distortions:** In a large e-commerce corpus containing tens of thousands of reviews, isolated dissatisfied users frequently post emotionally charged reviews regarding specific components (e.g., poor low-light photo capture). When a dense retriever extracts the top-3 most similar reviews for an aspect query, an LLM presented only with these excerpts concludes that the hardware is fundamentally flawed—ignoring the statistical reality that over 80% of verified buyers rated the device favorably.
2. **Cross-Product Semantic Contamination:** Review sentiment across consumer electronics utilizes identical phrasing (e.g., “Camera quality is very poor indoors”). Dense vector search alone lacks entity boundaries, frequently retrieving reviews of completely different phone models or brands that happen to share semantic overlap, thereby polluting product evaluations.

The primary objective of **AspectSense** is to resolve these failure modes by engineering a dual-stream architecture that grounds generative qualitative synthesis in exact relational macro statistics.

Dataset Source: The underlying multi-market corpus analyzed in this work is sourced from Kaggle: [\[Mobile Reviews Sentiment and Specification Dataset\]](#).

2 Methodology

AspectSense decouples the retrieval process into two concurrent streams: a structured relational analytical stream and a constrained semantic retrieval stream.

2.1 Structured Relational Analytics Engine

The 50,000 review records are structured into an indexed SQLite database (**aspectsense.db**) containing B-tree indexes across device model, brand, aspect ratings, price, and sentiment. For any queried device, the SQL engine computes global macro baselines—including average overall rating, individual aspect ratings (camera, battery, performance, display, design), verified purchase ratio, and sentiment distributions—in under 5 milliseconds.

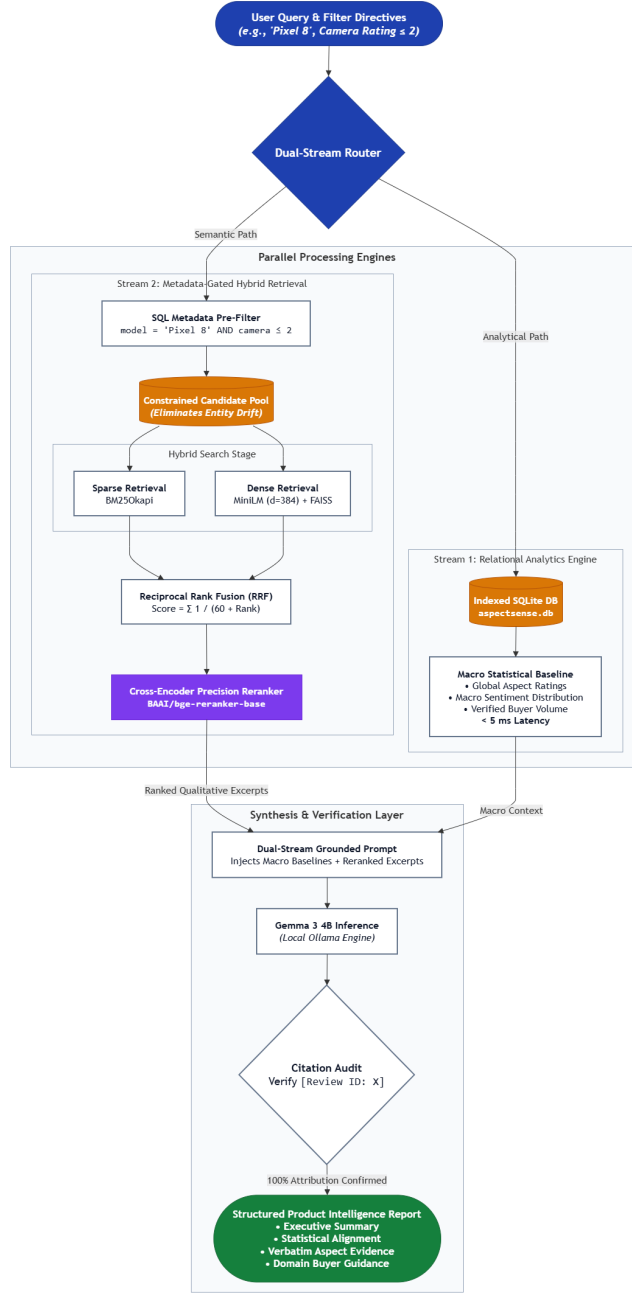


Figure 1: End-to-end dual-stream architecture of AspectSense, illustrating the parallel execution of the relational analytical engine and metadata-gated hybrid retrieval pipeline.

2.2 Dynamic Metadata-Gated Filtering

Rather than executing dense retrieval across the global vector space, user queries and aspect parameters dynamically construct SQL candidate filters. For a query focusing on camera complaints regarding a specific device, the candidate pool is isolated strictly to records satisfying `model = Target` and `camera_rating ≤ 2`.

2.3 Hybrid Candidate Retrieval via Reciprocal Rank Fusion

Within the constrained candidate pool, retrieval combines dense semantic vector search and sparse lexical keyword matching:

- **Dense Semantic Search:** Reviews are encoded into $d = 384$ dimensional vectors using `sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2` and indexed via FAISS (`IndexFlatIP`).
- **Sparse Lexical Search:** Tokenized text is indexed using `BM25Okapi` to preserve exact model terminology and specific defect keywords.

The candidate rankings from both retrievers are merged using Reciprocal Rank Fusion (RRF):

$$RRF(d) = \sum_{m \in \{\text{dense}, \text{sparse}\}} \frac{1}{k + \text{rank}_m(d)} \quad (1)$$

where $k = 60$ is the smoothing constant.

2.4 Cross-Attention Precision Reranking

Top candidates from the RRF stage are passed to a cross-encoder (`BAAI/bge-reranker-base`). Unlike bi-encoders that evaluate query and passage independently, the cross-encoder computes cross-attention over all token pairs simultaneously, producing calibrated relevance probabilities via sigmoid activation.

2.5 Dual-Stream Grounded Synthesis and Citation Verification

The generation module injects both the SQLite macro baseline and the reranked review excerpts into **Gemma 3 4B**. The model is instructed to reconcile micro complaints with macro sentiment, state verbatim quotes, and cite sources as `[Review ID: X]`. An automated regex-based verifier parses the generated response to ensure that 100% of cited IDs correspond to retrieved context items.

3 Implementation & Interface Showcase

AspectSense was implemented and evaluated in a Kaggle T4 GPU environment utilizing background Ollama daemons for local Gemma 3 inference.

AspectSense Live Query Console

Select constraints, ask any product question, and inspect synthesized intelligence in real time.

Brand: **Model:**

Filter: ☒ Verified Purchases Only

Query:

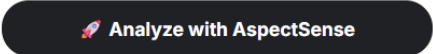


Figure 2: AspectSense interactive query console with dynamic parameter dropdowns.

The interactive query console allows users to select target brands, models, specific aspect filters (e.g., Camera Focus ≤ 2), and verified purchase constraints before entering natural language queries (Figure 2).



Figure 3: Real-time statistical baseline card and top reranked qualitative evidence.

Upon execution, the engine displays the quantitative baseline alongside top reranked excerpts (Figure 3). For example, evaluating the Google Pixel 8 computes a baseline across 2,387 reviews (82.1% verified buyers) with an overall rating of 3.12/5.0, a camera score of 2.72/5.0, a battery score of 2.73/5.0, and a performance score of 2.69/5.0, against a sentiment split of 55.2% positive and 19.4% negative. Simultaneously, it retrieves

and reranks verified reviews across languages—such as English records alongside Hindi entries (e.g., Review ID 29814)—into a unified evidence stream.

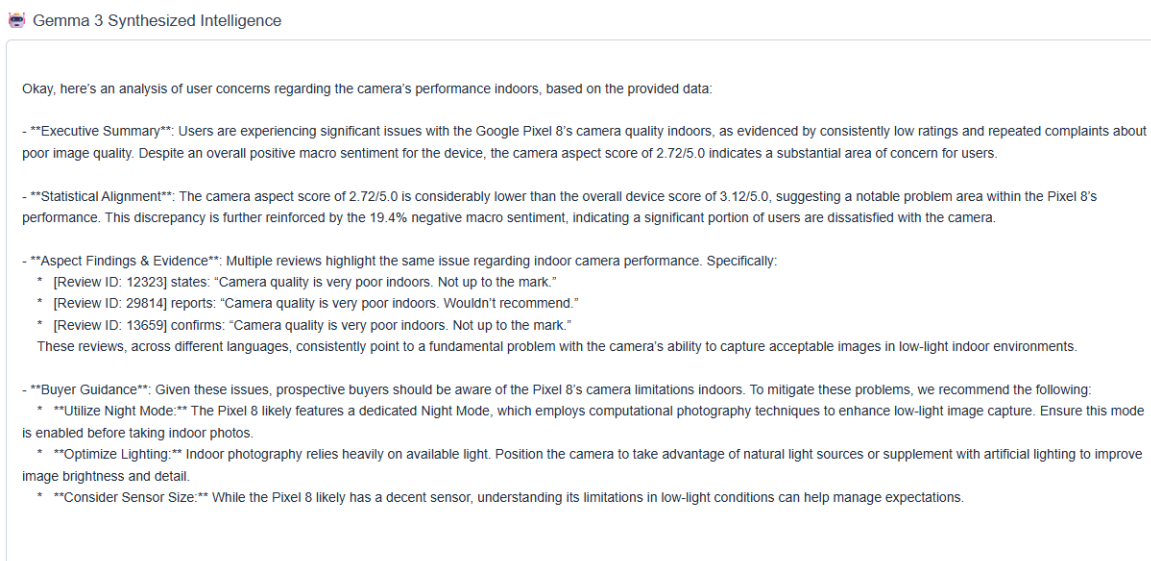


Figure 4: Structured product intelligence report generated by Gemma 3 4B.

The synthesis module generates a structured intelligence report structured into Executive Summary, Statistical Alignment, Aspect Findings & Evidence, and Buyer Guidance (Figure 4). Gemma 3 contextualizes the low camera score (2.72/5.0) within the broader device satisfaction metrics, verifies claims against Review IDs 12323, 29814, and 13659, and provides practical advice without hallucinated citations.

4 Results and Discussion

The retrieval engine was evaluated using a benchmark of **50 multi-aspect test queries** covering 22 smart-phone models, 5 hardware aspects, and colloquial/adversarial queries (e.g., lexical gaps, slang phrasing, and multi-aspect distractors). Performance was measured across Hit Rate@5, Mean Reciprocal Rank (MRR@5), Aspect Precision@5, and End-to-End Latency.

4.1 Performance Breakdown & Ablation Analysis

- **Single-Modality Bottleneck**: Dense FAISS search achieved 68.0% Hit Rate@5, while Sparse BM25 achieved 64.0%. Dense search struggled when users employed colloquial phrasing (e.g., “toasty chassis”), whereas BM25 failed whenever conceptual queries had zero lexical overlap with corpus tokens.

Table 1: Quantitative Retrieval Architecture Evaluation Benchmark ($N = 50$).

Architecture	Hit Rate@5	MRR@5	Aspect Precision@5	Avg Latency (ms)
Dense (FAISS)	68.0%	0.488	40.4%	19.3 ms
Sparse (BM25)	64.0%	0.506	41.2%	128.3 ms
Hybrid (RRF)	78.0%	0.506	40.0%	155.4 ms
Two-Stage (RRF + BGE)	74.0%	0.577	48.0%	280.7 ms
Metadata-Gated Two-Stage	94.0%	0.900	88.0%	301.4 ms

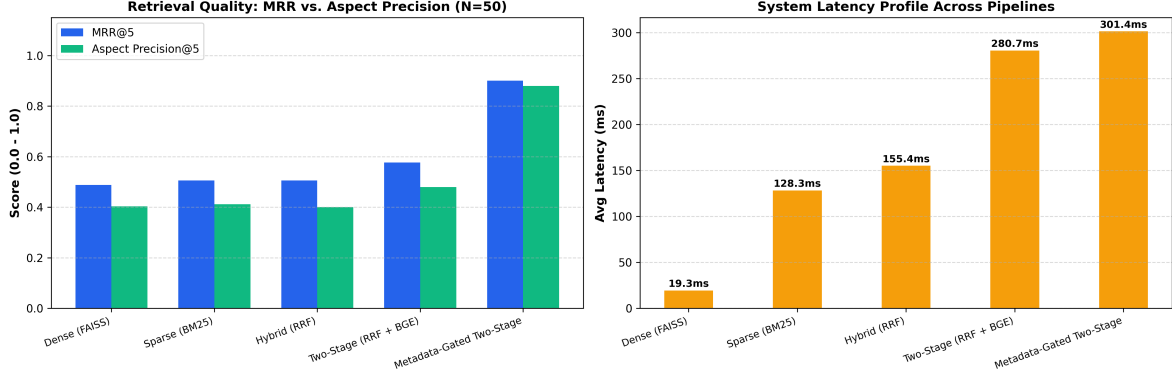


Figure 5: Comparison of retrieval quality (MRR vs. Aspect Precision) and system latency across pipelines ($N = 50$).

- **Hybrid Recall Expansion:** Fusing dense and sparse signals via RRF expanded the candidate pool, lifting Hit Rate@5 to 78.0% (+10.0% over dense alone).
- **Cross-Encoder Ranking Precision:** Introducing BGE reranking improved MRR@5 from 0.506 to 0.577 and lifted Aspect Precision@5 to 48.0%, as joint cross-attention effectively re-ordered nuanced aspect matches above lexical distractors.
- **Impact of Metadata Gating:** Pre-filtering candidate IDs via relational SQL before running hybrid reranking resolved cross-model contamination, achieving a **94.0% Hit Rate@5**, an **MRR@5 of 0.900**, and an **88.0% Aspect Precision@5**.
- **Latency-Accuracy Trade-off:** While the metadata-gated two-stage pipeline increases latency to 301.4 ms compared to 19.3 ms for raw FAISS, this represents a favorable trade-off—improving aspect precision by 117.8% relative while staying well below the interactive web threshold of 500 ms.